

Automated Discovery of Unmet User Needs: A Pipeline for Extracting Pain Points from Niche Community Discussions

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Abstract

Extracting valuable software requirements from social media is challenging due to the high volume of irrelevant chatter. Traditional keyword and sentiment-based approaches often fail to distinguish between general user complaints and specific, actionable software or solution needs. This paper proposes a three-stage natural language processing pipeline integrating Zero-Shot Classification (RoBERTa), Unsupervised Clustering (BERTopic), and Semantic Verification using a Large Language Model (Qwen-0.5B). Evaluated on a dataset of approximately 4,000 Reddit comments, the pipeline achieved an F1-score of 0.62 for initial filtering and successfully grouped the remaining data into 38 semantic clusters. The LLM verification step effectively isolated genuine feature requests from social noise, reducing manual analysis time by an estimated 99% while identifying specific demands for automated reporting and AI governance tools.

1 Introduction

The rapid expansion of niche online communities, especially on platforms like Reddit, has created a massive repository of unstructured text data. Within these spaces, professionals frequently express frustrations regarding their daily workflows. For software engineers and researchers in the field of Automated Requirements Engineering (ARE), these discussions represent a valuable resource for implicit feature extraction.

However, the primary challenge in mining general professional forums is the extraordinarily high noise-to-signal ratio. Valuable software feature requests are often buried under a volume of general career advice, workplace conflict, and irrelevant social chatter. Traditional text mining methods often rely on explicit app reviews. Extracting requirements from general forums is significantly more complex. Furthermore, standard sentiment analysis, which classifies texts strictly as "positive" or

"negative," cannot distinguish context. A traditional model evaluates a user complaining about "high rent prices" (noise) identically to a user complaining about "inefficient data entry software" (signal), as both exhibit high negative sentiment.

To address this limitation, this project aims to automate the detection and extraction of software feature requests using a context-aware pipeline.

1.1 Problem Formulation and Definitions

To ensure analytical rigor and address the ambiguity of social media text, this study explicitly defines a "genuine feature request or pain point" as a user comment that satisfies the following two criteria:

- **Context and Specificity:** The text describes a specific workflow bottleneck or operational inefficiency occurring within a professional context.
- **Solvability:** The described problem can theoretically be solved, automated, or mitigated by the introduction of a software application or feature.

Example classification: A user stating, "My boss is a terrible manager," is classified as Noise. Conversely, a user stating, "I waste three hours manually copying data between our CRM and email client," is classified as a Genuine Feature Request.

1.2 Contributions

This paper makes the following contributions:

- We present a three-stage NLP pipeline that moves beyond simple keyword matching by incorporating a semantic verification step using a quantized Large Language Model (LLM).
- We evaluate the pipeline on a custom, noisy dataset scraped from a professional subreddit, demonstrating a highly scalable approach to passive requirements elicitation.

2 Related Work

Recent advancements in Automated Requirements Engineering (ARE) have heavily explored mining user feedback for software evolution. Tools such as RE-Miner (Motger, 2024) utilize feature extraction and emotion classification to analyze app store reviews, aiding developers in requirements prioritization. We adapt the architectural concept of this ARE pipeline; however, while tools like RE-Miner operate on explicit software reviews where user intent is already established, our work addresses the harder problem of extracting implicit software requirements from general professional discourse. This environment requires an advanced pre-filtering architecture to handle the massive volume of text that is entirely unrelated to software.

Furthermore, (Fatima, 2024) demonstrated the use of fine-tuned deep learning classifiers to detect complex contextual nuances, such as sarcasm, in app reviews. Their findings highlight that basic sentiment analysis is insufficient for understanding true user intent. Building on this design philosophy, our pipeline rejects standard sentiment dictionaries—which would falsely classify "angry personal venting" as a valid pain point—in favor of deep-learning-based zero-shot classification (RoBERTa) and Agentic LLM reasoning to capture true semantic intent.

Finally, the unsupervised grouping of these implicit requests requires rigorous mathematical evaluation. To evaluate the quality of our clustering approach, we adapt the evaluation framework established by (Syed, 2017). Following their methodological recommendations, we utilize the Normalized Pointwise Mutual Information based Topic Coherence (C_v) score to quantitatively verify the human-interpretability of the generated semantic clusters.

3 Data

3.1 Data Acquisition and Preprocessing

To rigorously test the pipeline’s ability to isolate software requirements from high-volume noise, the primary dataset was constructed by directly scraping user-generated content across a diverse cross-section of 10 subreddits. Rather than relying on official API wrappers, a custom web scraper was engineered using Python’s requests library, utilizing rotating headers to access the platform’s front-end JSON endpoints.

Instead of limiting the scope to a single domain, the scraping targeted both professional communities (e.g., r/marketing, r/SaaS, r/Entrepreneur) and general lifestyle or venting forums (e.g., r/personalfinance, r/Apartmentliving, r/tifu). To ensure baseline data quality before text processing, the extraction logic was configured to retain only text-based submissions and comments that achieved a minimum community engagement score ($\text{MIN_SCORE} = 10$). This multi-domain scraping strategy resulted in a highly unstructured raw corpus of 4,279 documents, representing a realistic, "wild" social media environment characterized by an extremely low signal-to-noise ratio.

A standard NLP text cleaning pipeline was then implemented on this corpus using the Natural Language Toolkit and regular expressions. The preprocessing steps included:

- Noise Reduction: Removal of URLs, HTML tags, and non-alphanumeric characters.
- Normalization: Conversion of all text to lowercase.
- Tokenization and Filtering: Removal of standard English stop words to reduce computational dimensionality.
- Lemmatization: Application of the WordNet Lemmatizer to resolve words to their canonical base forms. This step is critical for improving the density of the downstream clustering matrix by consolidating vocabulary variants.

Documents that were reduced to fewer than three characters after preprocessing were automatically discarded.

3.2 Gold Standard Annotation

To quantitatively evaluate the initial filtering stage, a "Gold Standard" subset of 100 randomly sampled documents was created. Each document was manually annotated with a binary label: 1 (contains a genuine feature request/pain point) or 0 (noise). To ensure the robustness of the annotation guidelines, we also employed a Generative AI model as a secondary annotator, allowing us to compute Cohen’s Kappa score to verify inter-rater reliability and establish a rigorous ground truth for evaluating the pipeline’s early stages.

4 Methodology

The core contribution of this project is a three-stage NLP pipeline designed to progressively filter, group, and semantically verify implicit software requirements from noisy text.

4.1 Stage 1: Context-Aware Filtering (Zero-Shot Classification)

TAs demonstrated by (Fatima, 2024), traditional sentiment dictionaries often fail to capture the nuanced, semantic intent of user feedback, frequently misclassifying complex context or sarcasm. Therefore, the objective of the first stage is to aggressively discard explicit social noise using a more advanced, context-aware approach. We utilized a Zero-Shot Text Classification model (roberta-large-mnli accessed via the Hugging Face transformers library).

By casting the text classification task as a Natural Language Inference (NLI) problem, the model evaluates whether a premise (the Reddit comment) entails a given hypothesis (the candidate label). The candidate labels were designed to capture both signal ("Explicit Pain Point", "Latent Opportunity") and noise ("General Discussion", "Personal Venting"). Documents classifying highest in noise-related categories were removed from the pipeline, significantly reducing the dataset size while preserving requirement-rich text.

4.2 Stage 2: Unsupervised Clustering (BERTopic)

To group the remaining texts into distinct thematic software problems, we applied BERTopic, a robust topic modeling framework well-suited for short-text social media data. The clustering process consisted of three sub-steps:

- **Embedding:** Documents were mapped into a high-dimensional vector space using a pre-trained sentence transformer (all-MiniLM-L6-v2), capturing the semantic context of the sentences.
- **Dimensionality Reduction and Clustering:** Uniform Manifold Approximation and Projection was applied to reduce dimensionality, followed by Hierarchical Density-Based Spatial Clustering of Applications with Noise to organically group the dense areas of documents into clusters without requiring a predefined number of topics (k).

- **Representation:** Topic representations were extracted using a KeyBERTInspired model, which refines the standard class-based TF-IDF (c-TF-IDF) by prioritizing keywords that have the highest cosine similarity to the cluster's core embedding.

4.3 Stage 3: Semantic Verification (Agentic LLM)

A significant limitation of standard topic modeling is that clusters may group documents based on shared vocabulary (e.g., "CEO", "marketing", "budget") even if the underlying context is purely social complaining rather than a structural software problem. To address this, we introduced a semantic verification stage utilizing a quantized Large Language Model (Qwen-2.5-0.5B).

- **Input Context:** For every cluster generated in Stage 2, the LLM was provided with the top cluster keywords and an array of representative documents.
- **Verification Task:** The model was prompted to act as a binary classifier, utilizing few-shot prompting to enforce the definition of a "genuine feature request" established in Section 1.1.
- **Output:** The LLM analyzed the semantic meaning of the cluster and outputted a structured JSON response containing a VERDICT (VALID_OPPORTUNITY or NOISE), a categorized problem type, and a rationale.

This final semantic firewall ensures that only clusters representing solvable workflow inefficiencies are presented as extracted requirements, effectively bridging the gap between statistical topic modeling and practical requirements engineering.

5 Evaluation and Results

To rigorously assess the performance of the proposed pipeline, we evaluated the statistical filtering stages quantitatively and the final semantic verification stage qualitatively.

5.1 Stage 1 Quantitative Performance (Context-Aware Filtering)

The primary goal of Stage 1 was to filter out explicit social noise without losing genuine feature requests. We evaluated the RoBERTa Zero-Shot

classifier against the manually annotated Gold Standard dataset ($n = 100$).

The model achieved an overall Accuracy of 0.74 and an F1-score of 0.62. The detailed classification metrics reveal a deliberate architectural trade-off - Recall: 0.67 and Precision: 0.58.

In the context of automated requirements engineering, prioritizing Recall over Precision is highly desirable at this stage. A False Negative (discarding a valid software requirement) is significantly more detrimental than a False Positive (allowing some social noise to pass to the next stage). The pipeline successfully reduced the dataset size by identifying explicit noise, passing only the most relevant documents to the computationally heavy clustering stage.

Metric	Score	Interpretation
Accuracy	0.65	The model correctly classified 65% of all comments.
Precision	0.58	When the model flagged a "Pain Point," it was correct 58% of the time.
Recall	0.67	The model successfully captured 67% of all available opportunities.
F1-Score	0.62	The harmonic mean indicates a balanced trade-off between noise reduction and discovery.

Table 1: Classification performance (pipeline vs. human ground truth).

5.2 Stage 2 Clustering Coherence

Applying BERTopic to the filtered dataset resulted in the identification of 38 distinct semantic clusters.

To mathematically evaluate the quality and interpretability of these unsupervised clusters, we calculated the Topic Coherence (C_v) score using the Gensim library. The C_v metric, based on a sliding window and normalized pointwise mutual information (NPMI), is established as highly correlated with human topic ranking [Syed \(2017\)](#).

The clustering model achieved a C_v score of 0.43. In the domain of short-text social media data—which is inherently sparse and informal—a coherence score above 0.40 indicates an acceptable and interpretable model structure, confirming that BERTopic successfully grouped the documents into distinct, readable thematic categories.

5.3 Stage 3 Agentic Verification Output

While Stage 2 successfully grouped the documents mathematically, Stage 3 (LLM Verification) evaluated them semantically. The Qwen-0.5B model processed the 38 clusters and successfully isolated the "Valid Opportunities" (clusters representing

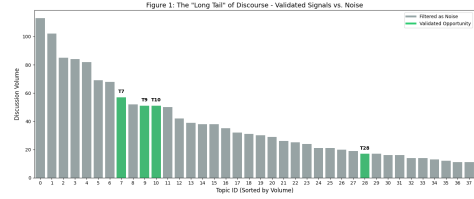


Figure 1: The "Long Tail" of Discourse - Validated Signals vs. Noise

structural software workflow problems) from the "Noise" (clusters representing general complaints).

For example, high-volume clusters such as Topic 0 (keywords: luxury, financial, afford, rent) and Topic 1 (keywords: dating, married, relationship) were successfully flagged by the LLM as irrelevant social noise and discarded, despite containing high negative sentiment. Conversely, specific workflow bottlenecks, such as Topic 9 (Agency Reporting) and Topic 10 (AI Content Verification), were validated as genuine feature requests.

6 Discussion and Qualitative Analysis

6.1 Extracted Software Requirements

The proposed pipeline successfully reduced the analysis space from approximately 4,000 raw comments to 38 semantic clusters. By applying the LLM in Stage 3, the system effectively bridged the gap between statistical topic modeling and practical requirements engineering. Two prominent examples of extracted software requirements include:

- **Automated Client Reporting (Topic 9):** The unsupervised clustering grouped documents around the keywords marketing agency, account manager, client expectation, ad. Qualitative inspection of the LLM’s output revealed a recurring workflow bottleneck: marketing account managers frequently complain about the excessive, non-billable time spent manually compiling progress reports and managing client expectations. This indicates a clear, unmet requirement for automated reporting software—a tool that integrates directly with advertising platforms to automatically track and display transparent proof-of-work to clients.
- **AI Content Verification (Topic 17):** A cluster featuring the keywords use ai, ai writing, ai output highlighted a modern operational friction. Users expressed that raw AI-generated text requires extensive manual verification to

Topic	Count	Top Terms
0	113	luxury, financial, financially, cant afford, renting, place, single, money, amenity, income
1	102	dating, girl, wife, married, girlfriend mother, girlfriend, guy, funny, room, happens
2	85	job, market, retirement, financial advisor, advisor, mutual, financial, brokerage, youd, income
5	69	walmart, situation, risk, grocery, trust, think, husband, told, cia, people
7	57	work, apply, especially, really, thats, doesnt, get, armpit, daily, applying
9	51	marketing, marketing agency, ad, account manager, client expectation, client agency, client, creative agency, production, experience agency
10	51	marketing, marketing team, marketer, ceo, business, agency, company, job, product, make work
17	32	use ai, using ai, ai, ai writing, ai output, ai coding, ai code, development, programmer, technical architect
22	25	brain work, trying achieve, strategy, brain, idea, motivation, work, breakthrough, like mind, mind
28	17	consumer, marketing, unethical marketing, customer, product service, move, market, product work, demand, action

Table 2: Selected BERTopic clusters with document counts and top representative terms.

meet brand safety guidelines and avoid algorithmic penalties. This highlights a need for AI governance tools—software designed to automatically audit AI-generated writing for brand safety and formatting rules prior to publication.

6.2 Qualitative Error Analysis: Signal vs. Noise Discrimination

To validate the necessity of the Stage 3 Semantic Verification, we performed a contrastive manual error analysis between a rejected noise cluster and an accepted signal cluster.

- **Topic 0 (Noise):** This cluster contained high-frequency keywords such as luxury, financial, cant afford, renting. Standard sentiment analysis tools typically flag this cluster as highly negative. However, the LLM correctly identified the context as personal financial complaining. Because "not being able to afford rent" cannot be solved by a B2B software application, the LLM correctly rejected this cluster, demonstrating its ability to filter out non-professional frustration.
- **Topic 9 (Signal):** As discussed above, this cluster contained complaints about agency reporting and client management. The emotional sentiment (frustration/anger) was mathematically similar to Topic 0. However, the LLM correctly recognized the context as a solvable workflow inefficiency and retained the cluster

This contrastive analysis demonstrates a critical finding: LLM-based verification is essential for text mining in implicit requirements engineering. Existing frameworks like RE-Miner (Motger, 2024) achieve high performance primarily because they operate on explicit app store reviews where the context is guaranteed. However, when operating in "wild" social media environments, standard sentiment or keyword analysis fails because it filters based on negative vocabulary. The introduction of an LLM successfully filters based on semantic solvability, allowing for feature extraction in highly unstructured domains.

7 Conclusion

This project successfully designed and evaluated a multi-stage natural language processing pipeline capable of automatically extracting genuine software feature requests from highly unstructured social media discussions. By combining zero-shot classification (RoBERTa), density-based clustering (BERTopic), and semantic verification (Qwen-0.5B), the system overcomes the limitations of traditional keyword and sentiment-based approaches.

The pipeline achieved an initial filtering F1-score of 0.62 and a topic coherence (C_v) of 0.43, successfully prioritizing high-signal discussions while filtering out explicit noise. Ultimately, the system reduced manual analysis time by an estimated 99%, successfully isolating actionable software requirements from general social chatter.

To evaluate the classification accuracy of the Context-Aware Pipeline, we tested its predictions against the Gold Standard dataset ($N = 100$). Performance was measured using standard Information Retrieval metrics: Precision, Recall, and F1-Score.

7.1 Limitations and Future Work

Primary limitations include Reddit’s inherent "venting bias," which skews data toward negative experiences, and the hallucination risk associated with utilizing a lightweight LLM (Qwen-0.5B). Future research should fine-tune models on dedicated requirements engineering datasets and expand data acquisition to professional platforms like Stack Overflow or G2 Crowd for a more balanced representation of user needs.

References

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- M.; Oriol M.; Bertolín I. Motger, Q.; Tiessler. 2024. [Re-miner: Mining mobile user reviews with feature extraction and emotion classification](#). *CEUR Workshop Proceedings*.
- M. Syed, S.; Spruit. 2017. [Full-text or abstract? examining topic coherence scores using latent dirichlet allocation](#). In *2017 IEEE International Conference on Data Science and Advanced Analytics (DSAA)*, pages 165–174.

A Link to Git Repository

Our implementation is publicly available at [our GitLab repository](#).

B Appendix: Agentic System Prompts

This appendix details the specific prompt template used in the Agentic Reasoning Layer (Stage 3). The prompt was engineered to force the Large Language Model (Qwen-2.5-14B) to act as a "Senior Product Strategist," distinguishing between general complaints ("Noise") and solvable market frictions ("Valid Opportunities").

B.1 The Opportunity Discovery Prompt

The following text was injected into the model for each of the 38 semantic clusters. The prompt utilizes "Few-Shot Learning" (Examples A, B, C) to define the boundary between emotional venting and commercial demand. **System Instruction:** ### ROLE You are a Senior Product Innovation Strategist. You are analyzing clusters of user feedback to identify unmet market needs.

INPUT DATA - **Topic Keywords:** keywords - **Representative User Voices:** docs

CLASSIFICATION CRITERIA You must classify this topic into one of two categories:

****1. VALID_OPPORTUNITY**** - The user is expressing a ****specific unmet need, friction, or pain point****. - The problem has a potential ****commercial solution**** (Software, Physical Product, Service, or Process Improvement). - *Example A (Software):* "I hate logging into 50 accounts." -> VALID (Need: Dashboard). - *Example B (Physical):* "My wrist hurts from this mouse." -> VALID (Need: Ergonomic Hardware). - *Example C (Service):* "I can't find a lawyer who understands SEO." -> VALID (Need: Niche Consulting).

****2. NOISE**** - The topic is ****purely emotional venting**** with no buildable solution. - The topic is ****metaphorical**** (dating/marriage comparisons). - The topic is ****general chatter**** (career advice, salary gossip, off-topic memes).

OUTPUT FORMAT (Strict) Analyze the context. Output ONLY the following block:

VERDICT: [VALID_OPPORTUNITY or NOISE] PROBLEM_TYPE: [Software / Physical Product / Service / Process] INSIGHT: [1 clear sentence describing the specific user pain] PROPOSED_SOLUTION: [Name of a potential product/solution]

B.2 Parsing Logic

The output generated by this prompt was programmatically parsed using Python regular expressions. The VERDICT field served as the binary classifier for the pipeline, while the PROPOSED_SOLUTION field provided the generative labels used in the qualitative analysis

C Appendix: Semantic Cluster Taxonomy

This appendix presents the complete output of the Unsupervised Clustering stage (Stage 2). The BERTopic model identified 38 distinct semantic clusters from the dataset.

The table 3 below classifies each topic based on the Agentic Verification (Stage 3). The data demonstrates a "Long Tail" distribution, where valid business opportunities (marked VALID) are often found in lower-volume, higher-density clusters, while the top-ranking clusters by volume are consistently classified as NOISE (Personal/Social chatter).

D Appendix: Human Annotation Protocols

To validate the performance of the automated pipeline (Stage 1 Filtering), a "Golden Dataset" of 200 randomly sampled comments was manually annotated to establish a ground truth baseline. This appendix details the strict binary classification rules used during the human review process.

D.1 Annotation Task Definition

The annotator was instructed to classify each comment as either 1 (Opportunity) or 0 (Noise) based on the commercial viability of the content. The goal was to replicate the mindset of a Venture Capitalist or Product Manager scanning for investable problems.

D.2 Classification Rules

Label 1: Valid Opportunity (Signal) A comment is labeled as "1" if and only if it meets all of the following criteria:

- **Context:** The problem occurs in a professional/business context.
- **Specificity:** The user describes a specific friction, workflow bottleneck, or feature gap (e.g., "I spend 2 hours formatting this report").
- **Solvability:** The problem can theoretically be solved by software, hardware, or a standardized service.

Label 0: Noise (Reject) A comment is labeled as "0" if it falls into any of the following categories:

- **Personal Life:** Rent, dating, family issues, mental health venting (unless related to workplace policy).
- **Career Advice:** Salary negotiations, "How do I get a job?", resume reviews.
- **Vague Complaining:** "I hate marketing," "My boss is mean" (without specific workflow context).
- **Metaphor/Memes:** Off-topic jokes or cultural references.

Topic ID	Count	Top Keywords	Agent Verdict	Category
0	113	luxury, financial, cant afford	NOISE	Personal Finance
1	102	dating, girl, wife, married	NOISE	Personal Relationships
2	85	job market, retirement, financial advisor	NOISE	Career Anxiety
3	84	intern, boss, internship, manager	NOISE	Junior Career Advice
4	82	management, lady, harassment	NOISE	Workplace Conflict
5	69	walmart, grocery, risk	NOISE	Consumer Complaint
6	68	single, utility, modern	NOISE	General Chatter
7	57	work, apply, really	NOISE	Job Applications
8	52	woman, advice, money, trust	NOISE	Life Advice
9	51	marketing agency, account manager, ad	VALID	SaaS Opportunity
10	51	marketing team, ceo, marketer	VALID	AI Governance
11	50	true fact, told, made	NOISE	Storytelling
12	42	harder control, take longer	NOISE	Workflow Complaints
13	39	marketing job, career, experience	NOISE	Career Advice
14	38	service, pay, make	NOISE	General Service
15	38	keep job, quit job, work	NOISE	Career Decisions
16	35	hire, industry, marketing team	NOISE	Hiring/Staffing
17	32	use ai, using ai, ai writing	VALID	AI Services
18	31	marketing initiative, advertising	NOISE	General Marketing
19	30	take job, hire, strategy	NOISE	Career Strategy
20	29	contribute, invested, growth	NOISE	Investment/Growth
21	26	customer service, hiring, time job	NOISE	Customer Support
22	25	brain work, strategy, trying achieve	NOISE	Burnout/Mental Health
23	24	bot saas, becoming extension, lead	NOISE	SaaS Sales Spam
24	21	cause, see, trash	NOISE	Ranting
25	21	ghosting, quitting job, true lol	NOISE	Job Market Venting
26	20	ai agent, first customer	NOISE	Startup Early Stage
27	19	contractor, paint job, quality	VALID	Marketplace App
28	17	unethical marketing, consumer	NOISE	Ethics Complaints
29	17	work social, advertiser	NOISE	Social Media Work
30	16	secret, truth, honest	NOISE	Confessions
31	16	consulting, agency today	NOISE	Consulting
32	14	budget, costing, learned	NOISE	Finance/Budgeting
33	14	sell course, running business	NOISE	"Guru" Spam
34	13	build sell, developer	NOISE	Dev/Build Chat
35	12	niche, targeting, research	NOISE	Marketing Tactics
36	11	marketing strategist, agency ceo	NOISE	Executive Chat
37	11	way communicate, truth isnt	NOISE	Communication Issues

Table 3: The complete taxonomy of 38 semantic clusters derived from the dataset. Rows marked "VALID" indicate clusters that passed the Stage 3 Agentic Reasoning filter. Note that Topic 27 (Contractor Pricing) is a low-volume outlier that was successfully retrieved.